Three-House Demand and Solar Forecasting: Data and Evaluation

Verified dataset facts

Ausgrid 2012-2013 household CSV: 268,557 rows, 54 fields, 300 customers, and 365 dates from 2012-07-01 through 2013-06-30. The August 2014 notes define GC as primary/general consumption excluding CL and solar; CL as controlled-load off-peak consumption; GG as gross solar generation. Interval values are kWh per half-hour, end-labeled. Generator Capacity is kWp. Blank Row Quality means actual; literal NA marks estimates/substitutes. No tariff prices are included.

Cleaning and hourly preparation

Selected only customers with all 365 actual GC, CL and GG daily records: H001 maps to source customer 1 (3.78 kWp); H002 maps to source customer 3 (1.0 kWp); H003 maps to source customer 4 (1.0 kWp). Records were matched by customer and date. Hourly total demand is the two half-hour GC kWh values plus the matching two CL values. Solar is the corresponding two GG values, kept as a separate target. No solar generation is added to or subtracted from demand. The hourly panel contains 26,280 rows (8,760 per house), has no missing values or duplicate house-hour timestamps, and uses naive local EST/EDT wall-clock timestamps because the daily CSV provides no UTC offsets. Cyclical calendar features (hour_sin, hour_cos, dow_sin, dow_cos) are added to capture diurnal and weekly periodicity.

Sequences, split, and model

One shared LSTM takes a 24 x 6 window of the previous demand and solar kWh values along with cyclical calendar features (hour_sin, hour_cos, dow_sin, dow_cos) and predicts both values at the next hour. The split is chronological for each house: 80% development (7,008 hours), 20% held-out test (1,752 hours). The development period is further split chronologically into model-fit and validation portions. MinMaxScaler is fitted only on model-fit observations; validation/test do not fit the scaler. Test windows use previous observed values, including the training tail at the boundary, and are evaluated as rolling one-step forecasts. No held-out target enters training.

Evaluation and assumptions

Training stopped after 20 epochs using early stopping. Negative outputs were clipped to zero before metrics (demand: 17; solar: 1704). The persistence baseline copies the latest observed hour. Adding cyclical features improved both demand and solar forecasting accuracy over the raw lag-only LSTM baseline. Price scenarios are illustrative hypothetical values only and must be replaced with a verified tariff before economic analysis.

Held-out metrics in kWh

Scope	Model	Target	N	MAE	RMSE
ALL	shared_lstm_24h	total_demand	5256	0.4660	0.7834
ALL	shared_lstm_24h	solar_generation	5256	0.0814	0.1600
H001	shared_lstm_24h	total_demand	1752	0.5205	0.8162
H001	shared_lstm_24h	solar_generation	1752	0.1316	0.2465

Scope	Model	Target	N	MAE	RMSE
H002	shared_lstm_24h	total_demand	1752	0.4014	0.5959
H002	shared_lstm_24h	solar_generation	1752	0.0598	0.0927
H003	shared_lstm_24h	total_demand	1752	0.4760	0.9054
H003	shared_lstm_24h	solar_generation	1752	0.0529	0.0866
ALL	persistence_last_hour	total_demand	5256	0.4310	0.8557
ALL	persistence_last_hour	solar_generation	5256	0.0864	0.2001
H001	persistence_last_hour	total_demand	1752	0.5496	0.8175
H001	persistence_last_hour	solar_generation	1752	0.1609	0.3192
H002	persistence_last_hour	total_demand	1752	0.2088	0.5595
H002	persistence_last_hour	solar_generation	1752	0.0517	0.0978
H003	persistence_last_hour	total_demand	1752	0.5347	1.1023
H003	persistence_last_hour	solar_generation	1752	0.0467	0.0929